

WORKSHEET

Longitude and time zones

You will need to use the 'Time zones map' to answer the questions in this worksheet. The map is available on NelsonNet.

- 1
 - a How is the time in Beijing different to the time in London?
 - b When it is 4 a.m. in London, what is the time in Beijing?
 - c Calculate the time in London when it is 6 p.m. in Beijing.
- 2 On the map, there are two locations that are 5 hours behind London. What are the two places?
- 3 In what location on the map is the time based on longitude of 135°E ?
- 4 What is the time difference between cities on the 120°W and 45°W meridians of longitude?
- 5
 - a Calculate the time difference between Johannesburg and Melbourne.
 - b Calculate the time in Melbourne when it is 9 a.m. in Johannesburg.
 - c What time is it in Johannesburg when it is midnight in Melbourne?
- 6 It is 12 noon on Monday in Auckland. What day and time is it in Tokyo, Bombay, Berlin and Honolulu?
- 7 The time in Vancouver in Canada is based on the 120°W meridian on longitude.
 - a What place on the map has the same time as Vancouver?
 - b When it is 2p.m. in Moscow, what time is it in Vancouver?
- 8 Caroline is a passenger on a ship. The ship's position coordinates are 15°S , 75°E .
 - a What city on the map has the same time as the ship Caroline is on?
 - b It is 8 a.m. on the ship. What time is it in Perth?
- 9 Paul caught a flight in Bombay at 6 pm Wednesday to fly home to Melbourne.
 - a What time was it in Melbourne when the flight left Bombay?
 - b The flight via Singapore took 14 hours. What time was it in Melbourne when the flight arrived?
- 10 Samantha is booking a flight to fly home to Perth from Johannesburg. The 9-hour-flight leaves Johannesburg at 10.25 p.m. Monday. Calculate the local time the flight is due to arrive in Perth.

Answers

- 1 **a** Beijing is 8 hours ahead
b 12 noon
c 10 a.m.
- 2 New York and Lima
- 3 Beijing
- 4 5 hours
- 5 **a** 8 hours
b 5 p.m.
c 4 p.m.
- 6 Tokyo 9 a.m. Monday
Bombay 5 a.m. Monday
Berlin 1 a.m. Monday
Honolulu 1 p.m. Sunday
- 7 **a** Seattle
b 4 a.m.
- 8 **a** Bombay
b 11 a.m.
- 9 **a** 11 p.m. Wednesday
b 1 p.m. Thursday
- 10 1.25 p.m. Tuesday